

What Actually Is AI?

60 Scenario-Based Multiple Choice Questions

Q1. A university librarian uses two systems: one that finds a book by exact catalog number typed in, and another that reads a student's request in plain language ("something on climate change for beginners") and suggests relevant titles it wasn't explicitly told about. Which system's behavior better reflects what AI actually means?

- A. The second system, because it interprets an open-ended request and produces a suitable response rather than matching an exact stored input
- B. Neither system, since a librarian is still available to help students in person
- C. The first system, because searching by catalog number is faster and more reliable for the librarian
- D. Both systems equally, since each retrieves information from the library's database

Q2. A phone's calculator app performs exact arithmetic every time it is given numbers and an operation. A separate app on the same phone predicts the next word you're likely to type based on your past messages. Which app is doing something closer to what AI actually is?

- A. Neither app, since phone apps only follow instructions written by programmers
- B. The word-prediction app, because it is inferring a likely outcome from patterns rather than computing one fixed correct result
- C. The calculator app, because arithmetic requires more precise internal logic
- D. Both apps equally, since each one processes input the user provides

Q3. A teacher tells a class, "Artificial Intelligence is just any computer program that does something impressive." A student disagrees. What is the strongest reason the student's objection is valid?

- A. AI must always involve a chatbot-style interface that a user can talk to
- B. AI programs are impressive only when they outperform a human at the same task
- C. AI can only run on very powerful, expensive hardware, unlike ordinary programs
- D. Being impressive is not the defining trait of AI; what matters is whether the system learned to produce outputs from data rather than following fixed, pre-written instructions

Q4. A small business owner says, "I built an Excel sheet with formulas that automatically total my monthly sales, so I built an AI tool." Is the business owner correct?

- A. No, because AI can only exist inside dedicated software, never inside Excel
- B. Yes, because the spreadsheet processes numerical data automatically without manual recalculation
- C. Yes, because Excel formulas can handle very large amounts of sales data at once
- D. No, because the spreadsheet only applies fixed formulas the owner wrote; it does not learn patterns or make predictions from data

Q5. An online store has a shipping-cost calculator that always charges a fixed rate based on the customer's entered zip code, following a pre-set rate table. A separate feature on the same site estimates delivery delay risk by learning from patterns in thousands of past deliveries. Which is the AI feature?

- A. Both features, because they are both parts of the same online store's checkout system
- B. The delivery-delay estimator, because it was built by learning from historical delivery data rather than a fixed rate table
- C. The shipping-cost calculator, because it directly affects what the customer pays
- D. Neither feature, since online stores only use basic database lookups

Q6. A payroll program calculates employee salaries using tax rules typed in by an accountant. If the tax law changes, someone must manually update the code. Why does this NOT count as AI, even though it automates a task?

- A. It only executes rules explicitly written by a person; it never learns patterns or adapts on its own from data
- B. It cannot be AI because payroll is a financial task rather than a technical one
- C. It cannot be AI because it produces the same output for the same input every time
- D. It only counts as AI once it is connected to the internet for real-time updates

Q7. A hospital uses software that flags a patient's file if their scheduled appointment is more than 30 days overdue, based on a rule set by administrators. A second hospital tool scans X-ray images and flags ones that resemble patterns seen in prior cases of a certain condition. Which tool demonstrates AI, and why?

- A. The appointment tool, because it directly affects patient scheduling and care
- B. The X-ray tool, because it was trained to recognize patterns in image data rather than following an administrator's explicit rule
- C. Neither tool, since hospital software must always be reviewed by a doctor first
- D. Both tools, because each one flags something automatically for hospital staff

Q8. Two apps both "recommend" a task to a user. App X always suggests the single oldest unfinished item in a to-do list. App Y studies each user's past completions and suggests whichever task the user is statistically most likely to finish next. Which app's recommendation behavior reflects AI?

- A. Both apps, since each one is described as making a recommendation
- B. Neither app, since to-do list software is inherently simple
- C. App X, since sorting by age of task is a more objective and fair method
- D. App Y, since its suggestion is derived from learned patterns in the user's own behavior rather than one fixed sorting rule

Q9. A smart thermostat lowers the temperature every day at exactly 10 PM because it was manually programmed with that schedule. A different thermostat learns from a household's actual patterns of being home or away and adjusts the temperature accordingly, even on days that don't match any fixed schedule. Which one displays intelligent, AI-like behavior?

- A. The second thermostat, because it adapts its behavior based on learned patterns rather than a fixed, unchanging schedule
- B. Neither thermostat, since temperature control is a basic household function
- C. Both thermostats, because each one automatically adjusts the temperature
- D. The first thermostat, because a consistent schedule is easier for a household to predict

Q10. A security camera turns on a light whenever its built-in motion sensor detects any movement, including a moving tree branch. A newer camera model distinguishes between a person, an animal, and blowing debris before deciding whether to send an alert. What makes the newer camera closer to AI?

- A. It uses a smaller, more energy-efficient motion sensor overall
- B. It interprets and classifies what caused the motion, rather than reacting identically to any detected movement
- C. It can be controlled remotely through a smartphone app
- D. It turns on the light faster than the older camera in low-light conditions

Q11. A vending machine dispenses a snack only when the exact correct coins are inserted, based on fixed mechanical rules. Could a vending machine ever be described as exhibiting AI-like intelligent behavior?

- A. No, because AI can never be applied to physical machines like vending machines
- B. Yes, because it responds differently depending on which button the customer presses
- C. Yes, because dispensing the correct snack for the correct payment already shows decision-making
- D. Only if it were redesigned to learn from data, such as predicting which snacks to restock based on patterns in past purchases

Q12. A customer service phone tree says "Press 1 for billing, Press 2 for support" and routes calls based only on the button pressed. A newer system listens to what the customer says in their own words and routes the call based on the likely intent behind it. Which shows AI-like intelligent behavior, and why?

- A. The phone tree, because it gives the customer clear, predictable options
- B. Both systems, since each one ultimately routes the call somewhere
- C. Neither system, because call routing is a basic telecommunications function
- D. The newer system, because it interprets varied, unstructured input to infer intent rather than matching one fixed keypress

Q13. A streaming service's engineer says, "Our recommendation engine is machine learning, but our whole platform is AI." What is the most accurate description of the relationship between the two terms?

- A. Machine learning is a newer, unrelated replacement for the older term 'AI'
- B. AI only refers to robots, while machine learning refers to all software systems
- C. Machine learning is one approach used to build AI systems; not all AI must use machine learning, but the recommendation engine described is a form of AI
- D. Machine learning and AI are two completely unrelated fields with no overlap

Q14. A retailer's team builds a system that studies years of past sales to predict how much inventory to order each month. A colleague insists, "That's not really AI, it's just machine learning." How should this claim be evaluated?

- A. The claim is correct, since AI must involve robotics or physical hardware
- B. The claim is misleading, since machine learning and traditional software are identical concepts
- C. The claim is misleading, since machine learning is itself one of the main ways AI systems are built, not something separate from AI
- D. The claim is correct, since machine learning only applies to inventory and sales forecasting

Q15. A student writes in an exam, "Machine learning means a computer that changes its own code while running." Why is this description inaccurate?

- A. Machine learning only happens on supercomputers, not desktop computers
- B. Machine learning involves adjusting internal numerical patterns based on data during training, not rewriting the program's own code
- C. Machine learning always requires the system to be connected to the internet
- D. Machine learning is completely automatic and requires no data of any kind

Q16. A weather app uses a machine learning model to estimate the chance of rain based on years of historical weather patterns. A colleague argues this can't be "real" AI because it just produces a percentage instead of a definite yes/no answer. Is this argument sound?

- A. No, producing a probability based on learned patterns is a hallmark of how many AI/ML systems reason, not a disqualifying feature
- B. Yes, weather forecasting is a physical science and can never involve AI
- C. No, but only because weather apps are updated more frequently than other software
- D. Yes, real AI must always give a definite, certain answer rather than a probability

Q17. A company spends several months and significant computing resources having a model study a large collection of customer support conversations before it is ever used by a real customer. What is this process called?

- A. Training — the stage where the system is exposed to data and adjusted to improve its future predictions
- B. Inference — the stage where the system responds to a live customer question
- C. Deployment — the stage where the finished system is released to users
- D. Debugging — the stage where programmers fix errors in the code

Q18. After the support chatbot above is released, a real customer types a question and instantly receives a reply. What is happening at that moment?

- A. Calibration — the system is being re-tuned using the customer's data
- B. Training — the system is learning from this specific customer's question
- C. Inference — the already-trained system is producing an output for new input, without permanently changing itself
- D. Compilation — the system's code is being converted for this one use

Q19. A manager asks, “If our AI answers a thousand customer questions today, will it be smarter tomorrow because of that?” What is the accurate answer, assuming no further training is done?

- A. Yes, but only if the customers rate their satisfaction with the answers
- B. No, because customer questions are too varied to be useful for improvement
- C. Yes, because every interaction automatically improves the system's core knowledge
- D. No, because answering questions is inference, which does not change the system's underlying learned patterns

Q20. A photo app took two years to build: one year collecting and labeling millions of images, and another year having the system repeatedly compare its guesses to the correct labels. Once released, it instantly classifies any new photo a user uploads. Which two stages does this describe, in order?

- A. Labeling, then compiling
- B. Inference, then training
- C. Training, then inference
- D. Testing, then debugging

Q21. Two companies try to build a system that predicts customer churn. Company A has ten years of detailed customer behavior records. Company B has only two months of incomplete records. All else being equal, why is Company A likely to end up with a more reliable AI system?

- A. Because having more data guarantees the system will never make an incorrect prediction
- B. Because AI systems learn patterns from the data they are trained on, and more complete, relevant data generally supports stronger, more reliable predictions
- C. Because Company A's computers are automatically faster at processing information
- D. Because ten years of data means the system requires no further updates ever

Q22. A voice assistant trained mostly on recordings of adult speakers in one accent performs noticeably worse when used by children or speakers with a different accent. What does this best illustrate about AI?

- A. Voice assistants cannot process any audio that isn't from an adult speaker
- B. AI systems always perform equally well across every type of user once released
- C. An AI system's performance is shaped by the data it was trained on, and it tends to perform worse on cases underrepresented in that data
- D. The issue only occurs due to a hardware limitation in the microphone

Q23. A hiring-screening tool is trained on ten years of a company's past hiring decisions. A reviewer worries that if those past decisions were biased, the AI tool might repeat similar patterns. What principle does this concern illustrate?

- A. An AI system's outputs reflect patterns present in its training data, including any biases contained in that data
- B. Bias can only appear in AI systems that use images, not text-based hiring records
- C. AI systems automatically correct any bias present in the data they are trained on
- D. AI tools are legally required to ignore past company decisions during training

Q24. A startup wants to build an AI tool to detect defective products on a factory line but has only 50 example photos of defects. An engineer warns this is likely too few. Why would that be a reasonable concern?

- A. 50 photos is too few because AI systems require at least one photo per factory worker
- B. AI systems generally need a sufficiently large and varied set of examples to learn reliable patterns; a very small dataset limits how well it can generalize
- C. The concern only applies to video data, not still photographs
- D. The concern is unreasonable, since AI systems perform identically regardless of data size

Q25. An app estimates how long a food delivery will take based on distance, traffic, weather, and thousands of past deliveries, then gives a specific number of minutes. A user assumes this number is a guaranteed fact. What is a more accurate way to understand that number?

- A. It is the model's prediction, an estimate based on patterns in past data, not a guaranteed or certain fact
- B. It is a guaranteed fact because it was generated by a computer program
- C. It is a fixed rule set once by the delivery company and never recalculated
- D. It is a random number chosen without any relation to the delivery's actual details

Q26. A credit application tool outputs “72% likely to repay on time” for one applicant. What does this output actually represent?

- A. A certain, guaranteed outcome for that specific applicant
- B. A model's prediction expressed as a probability, based on patterns learned from data about past applicants
- C. The exact percentage of the applicant's income that will go toward the loan
- D. A legally binding score that cannot be reviewed by a human

Q27. A real-estate app predicts a house's sale price. Two nearly identical houses on the same street get noticeably different predicted prices. A confused homeowner assumes the app made an error. What is the most likely explanation?

- A. The app must be broken, since identical inputs always produce identical outputs in AI
- B. The homes must be located in two different unrelated markets
- C. Real-estate prediction apps are incapable of using property condition as a factor
- D. The model is weighing subtle differences in the data — such as condition, past sales nearby, or listing details — that don't seem obvious to a person

Q28. A weather-prediction model says there is an “80% chance of rain tomorrow,” and it does not rain. A user claims the model was simply wrong. Is this a fair conclusion?

- A. Yes, because weather predictions must always be stated as certainties, not probabilities
- B. No, because rain predictions are never based on probability in the first place
- C. Not necessarily, since an 80% chance still allows for a 20% chance of no rain; a single outcome doesn't prove the prediction process was flawed
- D. Yes, any prediction that doesn't match reality is automatically a failed model

Q29. A banking app flags a transaction as suspicious because it differs sharply from a customer's typical spending location, amount, and time of day, even though no single rule was explicitly broken. What is this system doing?

- A. Recognizing deviations from learned patterns in the customer's normal behavior
- B. Randomly selecting a small percentage of transactions to flag
- C. Applying a fixed list of banned merchants and transaction amounts
- D. Following a rule that all transactions above a certain fixed amount are suspicious

Q30. A photo-sorting app automatically groups a user's vacation photos by the people who appear in them, without the user ever labeling anyone. What ability does this most clearly demonstrate?

- A. Randomly assigning photos to groups based on the time they were taken
- B. Recognizing recurring visual patterns (faces) across many images and grouping based on similarity
- C. Following a strict folder structure created by the user beforehand
- D. Reading metadata tags that the camera manually attached to each photo

Q31. A quality-control camera on a factory line is trained on thousands of images of acceptable and defective bottles. It later flags a bottle with a crack pattern it has never seen in exactly that shape before. How is this possible?

- A. It must have been shown that exact crack pattern before, or it could not flag it
- B. It randomly flags a certain percentage of bottles regardless of appearance
- C. It generalizes from patterns learned across many examples, so it can recognize a similar-but-new defect rather than only exact matches
- D. A human operator manually reviews and flags every bottle in real time

Q32. A spam filter blocks a newly worded scam email it has never encountered before, even though the exact wording differs from any email in its training data. What best explains this?

- A. The filter recognized patterns similar to spam characteristics it learned from many prior examples, not an exact match to one specific email
- B. The filter contains a complete list of every possible scam sentence ever written
- C. The filter blocks all emails from unfamiliar senders regardless of content
- D. The email service manually reviewed and pre-approved the block before sending

Q33. A marketing employee types a short description into a tool and receives a brand-new product image that has never existed before, generated to match the description. What category of AI is this tool an example of?

- A. Data-retrieval software, since it fetches an existing stock photo
- B. Predictive AI, since it forecasts what the product will look like in the future
- C. Rule-based automation, since it follows a fixed image template
- D. Generative AI, since it produces new content rather than only classifying or predicting a label for existing content

Q34. A teacher uses a writing assistant to generate an entirely new short story based on a one-line prompt. A student asks, "Did it find this story somewhere online?" What is the most accurate answer?

- A. No, because generative AI tools cannot produce anything related to storytelling
- B. No, it generated new text by predicting a likely, coherent continuation based on patterns learned from a huge amount of text, not by retrieving an existing story
- C. Yes, but only because the prompt matched an existing story's title exactly
- D. Yes, it searched a database of published stories and copied the closest match

Q35. A design tool can generate dozens of unique logo concepts from a short text prompt, no two exactly alike. What underlying capability makes this generative behavior possible?

- A. The system was trained on large amounts of visual data and learned patterns it can recombine to produce new, varied outputs
- B. The tool duplicates the user's own past uploads with small color changes
- C. The tool asks a remote team of human designers to create each logo on demand
- D. The tool stores a large library of pre-made logos and randomly selects among them

Q36. A student uses a generative AI tool to write an essay and submits it without checking any of the facts, assuming everything stated must be accurate because it reads fluently. What risk does this scenario best illustrate?

- A. Generative AI can produce fluent, confident-sounding text that is factually incorrect, so its output still needs to be verified
- B. This risk only applies to spoken AI assistants, not text-generation tools
- C. Fluent writing from generative AI is always guaranteed to be factually accurate
- D. Generative AI tools are legally required to fact-check their own output before display

Q37. A navigation app can calculate optimal driving routes extremely well but cannot hold a conversation, write an essay, or recognize a photo. What term best describes this kind of AI?

- A. Symbolic AI — a system based entirely on hand-written logical rules
- B. General AI — a system capable of performing any intellectual task a human can
- C. Generative AI — a system that produces brand-new creative content
- D. Narrow AI — a system designed and trained to perform one specific task or a limited set of related tasks well

Q38. A company claims its new assistant “can do literally anything a human professional in any field can do, with no additional training.” Why should this claim be treated with skepticism?

- A. The claim is reasonable, since narrow AI and general AI are simply two names for the same thing
- B. Current AI systems are narrow, excelling at specific trained tasks; a system with fully general, human-equivalent ability across every domain does not yet exist
- C. The claim is reasonable, since most modern AI systems already qualify as general intelligence
- D. The claim should be doubted only because such assistants are always too expensive

Q39. A chess-playing AI can defeat nearly every human grandmaster but performs poorly if asked to plan a household budget, despite both being reasoning tasks. What does this best illustrate?

- A. The chess AI is a general AI system, since chess itself is an extremely difficult task
- B. The chess AI must be malfunctioning, since true intelligence should transfer to any task
- C. The chess AI is a narrow system, excelling within its specific trained domain but not transferring that skill to unrelated tasks
- D. Budgeting is inherently impossible for any computer system to assist with

Q40. A student argues, “Since my phone's AI assistant can set alarms, answer questions, and translate languages, it must be a general AI.” What is the best response to this claim?

- A. Handling several separate trained tasks within a limited scope is still narrow AI; general AI would require flexible, human-like reasoning across essentially any domain, not just several pre-built features
- B. The claim is correct, since performing more than one task automatically qualifies as general AI
- C. The claim is irrelevant, since general AI and narrow AI apply only to robots, not phones
- D. The claim is correct, since any AI available on a consumer smartphone is by definition general

Q41. A ride-hailing app adjusts prices in real time based on current demand, driver availability, and historical patterns in a given area. Which aspect of this system most clearly reflects AI in action?

- A. The price changes are displayed instantly on the user's smartphone screen
- B. The price is generated by analyzing patterns in historical and current data rather than a single fixed rate
- C. The app requires an internet connection to function at all
- D. The app allows the driver to accept or decline the trip request

Q42. A translation app converts a spoken sentence in one language into natural-sounding speech in another language almost instantly, without a human translator present. What makes this a real-world AI application?

- A. It works because a human translator is monitoring the app remotely in real time
- B. It was trained on large amounts of paired language data to learn how meaning maps between languages, rather than using a fixed word-for-word dictionary
- C. It works because the words in each language are stored as exact matching pairs
- D. It works because the app simply plays a pre-recorded phrase for common sentences

Q43. A photo app on a smartphone automatically tags photos as “beach,” “birthday,” or “pet” without any manual input from the user. Which real-world AI category does this best represent?

- A. Text generation, since the app is writing new descriptive captions from scratch
- B. Data encryption, since the app is protecting the photos from unauthorized access
- C. Speech recognition, since the app is interpreting spoken commands from the user
- D. Image recognition, since the app is classifying photo content based on learned visual patterns

Q44. A car's driver-assistance system detects a pedestrian stepping into the road and applies the brakes automatically, without the driver reacting in time. Which describes why this is an AI-driven capability rather than simple automation?

- A. The system must interpret varied, unpredictable visual and sensor data in real time to distinguish pedestrians from other objects, rather than follow one fixed trigger
- B. The system is AI-driven simply because it involves a moving vehicle
- C. The system is AI-driven because it operates without an internet connection
- D. The system is AI-driven because braking is a mechanical action

Q45. A news article claims, “The company’s AI chatbot understands and genuinely cares about its customers’ feelings.” Which statement best evaluates this claim?

- A. The claim is accurate, since customer service is inherently an emotional task
- B. The claim is accurate, since any chatbot that responds kindly must feel real empathy
- C. The claim overstates what is happening; the system predicts responses that sound empathetic based on patterns in training data, without genuinely experiencing feelings
- D. The claim cannot be evaluated because chatbots are never trained on customer data

Q46. A relative insists, “AI systems are always 100% objective and free of bias because they’re just math.” Why is this a misconception?

- A. AI systems learn from data created by people, so they can reflect and even amplify biases present in that data
- B. This is a misconception only for AI systems built outside of large companies
- C. This is correct, since bias can only occur when a human directly makes a decision
- D. This is correct, since mathematical processes cannot be influenced by outside factors

Q47. A colleague says, “If the AI gave a confident, detailed answer, it must be correct.” Why is this reasoning flawed?

- A. An AI system’s tone of confidence reflects a learned writing style, not a built-in check on whether the content is actually accurate
- B. This reasoning is sound, since confident answers are always double-checked before being shown
- C. This reasoning is sound, since AI systems only sound confident about topics they know perfectly
- D. This reasoning is flawed only when the AI is answering a math question

Q48. A friend says, “AI systems think and reason exactly the way the human brain does, just faster.” What is the most accurate response?

- A. This is accurate, since AI systems are built by directly copying brain scans
- B. This is inaccurate only for AI systems used in scientific research
- C. AI systems use statistical patterns learned from data to produce outputs; this process differs fundamentally from biological human thought, even when results look similar
- D. This is accurate, since faster processing is the only real difference between the two

Q49. A student asks an AI writing tool to summarize a textbook chapter and it does so clearly and accurately. The same student then asks it to guarantee that a specific stock will rise in value next week, and it gives a confident-sounding answer. What should the student conclude?

- A. Since it summarized well, its stock prediction must be equally reliable
- B. AI can be strong at tasks like summarization while being unreliable at tasks requiring certainty about unknowable future events, regardless of how confident the answer sounds
- C. AI tools are incapable of attempting a task like summarization or forecasting
- D. Confident phrasing on the stock question proves the prediction is well-researched

Q50. A company wants an AI system to independently decide whether to approve high-risk loans with zero human oversight, citing that “the AI can handle it alone.” Why might this plan be risky based on what AI can and cannot do?

- A. AI systems can make errors, reflect biased patterns from training data, and lack judgment about context, so unchecked high-stakes decisions carry real risk
- B. The plan is risky solely because loan approval must legally involve paper documents
- C. The plan is only risky because AI systems require a fee for every decision made
- D. AI systems are incapable of processing any financial information at all

Q51. A parent believes an AI tutor “knows” whether their child truly understands a concept, the same way a human teacher would sense it. What is a more accurate description of what the AI tutor is actually doing?

- A. It uses the exact same biological process a human teacher uses to sense understanding
- B. It is estimating understanding based on patterns in the student's answers and interactions, not perceiving understanding the way a person would
- C. It is reading the child's mind through the device's camera and microphone
- D. It has no way of assessing a student's performance in any form

Q52. A user is impressed that an AI image tool can generate a photorealistic picture of “a cat wearing a spacesuit on Mars,” something that has never existed. Does this mean the AI truly understands cats, spacesuits, or Mars the way a person does?

- A. Yes, because photorealism can only be achieved through genuine comprehension
- B. Yes, generating a coherent new image is only possible with full human-like understanding
- C. No, because the AI image tool did not actually create a new image at all
- D. Not necessarily; it can combine learned visual patterns into a new image without possessing human-like understanding or awareness of what those things are

Q53. An AI-powered resume screener rejects a highly qualified candidate because their resume used an unusual format the system wasn't trained to recognize well. What limitation does this best illustrate?

- A. This limitation only occurs in systems that have never screened a resume before
- B. AI systems can perform poorly on inputs that differ significantly from the patterns they were trained on
- C. AI systems always intentionally reject the most qualified candidates
- D. AI systems cannot process resumes in any format under any circumstances

Q54. A customer support chatbot gives a fluent, detailed, but completely incorrect answer about a return policy that doesn't actually exist. What limitation of many AI systems does this reflect?

- A. This limitation applies only to chatbots that use a text-based interface
- B. This only happens when the chatbot's internet connection is interrupted
- C. Such systems can produce confident, fluent responses that are factually wrong, since fluency and accuracy are not the same thing
- D. Such systems are physically incapable of producing any incorrect statement

Q55. An AI-based translation tool struggles significantly with a rare regional dialect that has very few written examples available anywhere. Why does this happen?

- A. Regional dialects are technically impossible for any software to process
- B. The issue is caused entirely by the user's microphone quality
- C. The system likely had little to no training data representing that dialect, limiting how well it learned to handle it
- D. The tool is deliberately programmed to ignore dialects to save computing power

Q56. A factory's defect-detection AI performs very well in bright lighting but frequently misses defects in dim lighting conditions it rarely saw during training. What does this demonstrate about AI systems in general?

- A. AI systems perform identically regardless of any change in real-world conditions
- B. Lighting has no relevant connection to how an image-based AI system operates
- C. The issue must be a hardware defect in the camera lens itself
- D. AI systems' performance can be sensitive to conditions that differ from their training environment

Q57. Engineers regularly review a customer-service AI's flagged conversations and adjust its training data to correct repeated mistakes. What role does this reflect?

- A. This process is only necessary for AI systems that have completely failed
- B. Ongoing human oversight and refinement remain an important part of maintaining and improving many AI systems after deployment
- C. Once an AI system is deployed, no further human involvement is ever needed
- D. This process shows the system is not really AI, since AI must be fully autonomous

Q58. A team of people labels thousands of images as "contains a stop sign" or "does not contain a stop sign" before a self-driving system is trained on them. What role are these people playing?

- A. They are providing the labeled data the system needs during training to learn to recognize stop signs
- B. They are testing the finished car for mechanical safety defects
- C. They are writing the underlying code that runs the self-driving system
- D. They are manually controlling the self-driving car's steering during operation

Q59. A content-moderation AI flags a borderline social media post, but a human moderator makes the final decision on whether to remove it. What does this workflow illustrate about many real-world AI systems?

- A. AI systems are never involved in any content-moderation process
- B. Human moderators only get involved when the AI system is completely offline
- C. This workflow proves the AI system serves no meaningful purpose in moderation
- D. Many AI systems are designed to assist human decision-making rather than fully replace human judgment, especially for sensitive decisions

Q60. A hospital's diagnostic AI suggests a possible condition based on scan results, but a licensed radiologist reviews the suggestion before informing the patient. Why is this human step generally considered important?

- A. It exists only to slow down the process for billing purposes
- B. It is unnecessary, since AI diagnostic suggestions are always guaranteed to be correct
- C. It provides an expert check on the AI's pattern-based suggestion, since the AI can make mistakes and lacks accountability and contextual medical judgment
- D. It is required only because hospital computers are too slow to display results directly